arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship
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arising from liquefied hydrogen cargo affecting persons on board, the environment, the structural strength or the integrity of the ship
are addressed. Consideration should be given to the hazards associated with properties of liquefied hydrogen and hydrogen gas, physical layout, operation and maintenance, following any reasonably foreseeable failure. For the risk assessment, appropriate methods, e.g. HAZID, HAZOP, FMEA/FMECA, what-if analysis, etc., should be adopted taking into account IEC/ISO 31010:2019 "Risk management Risk assessment techniques"7) and SAE ARP 5580 -2001 "Recommended failure modes and effects analysis (FMEA) practices for non-automobile applications"9).
General
(see 4.1)
A-24 Relief valve sizing should be undertaken for the most onerous scenario. The evaluation should include the fire scenario and should consider the resulting magnitude of the heat flux on the cargo containment system.
High pressure
hazard
(see 4.8)

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should consider the resulting magnitude of the heat flux on the cargo containment system.
High pressure
hazard
(see 4.8)
A-25 A filling limit exceeding 98% at reference temperature should not be permitted.
High pressure
hazard
(see 4.8)
A-26 Bolted flange connections of hydrogen piping should be avoided where welded connections are feasible.
Susceptibility to
leakage (see
4.4.2)
A-27 Due consideration should be given to the invisible nature of hydrogen fire from the viewpoint of safety of ships and especially personnel in case of fire.
Fire hazard
(see 4.7.1)
3 EXPLANATION ON GENERAL REQUIREMENTS

3.1 Properties of liquefied hydrogen
The application of general requirements in the Code for liquefied hydrogen has been considered based on a comparison study on the physical properties of liquefied hydrogen and LNG. LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate
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and LNG. LNG and liquefied hydrogen are cryogenic liquids, non -toxic, and generate flammable high pressure gas. For reference, table 3 shows the comparison of physical properties of hydrogen and methane, the major component of LNG.

Table 3: Comparison of physical properties of Hydrogen and Methane

	Hydrogen	Methane	References
Boiling temperature (K)*	20.3	111.6	ISO1), Annex A, Table A.3
Liquid density (kg/m3)*	70.8	422.5	ISO1), Annex A, Table A.3
Gas density (kg/m3)**	(Air: 1.198) 0.084	0.668	NIST RefProp10)
Viscosity (g/cms x 10-6)			
Gas			
Liquid			
	8.8		
	13.49		
	10.91		
	116.79		
	NIST RefProp10)		
	NIST RefProp10)		
Flame temperature in air (°C)	2396	2230	Calculated using Cantera and GRI 3.0 mechanism
Maximum burning velocity (m/s)	3.15	0.385	Calculated using Cantera and GRI 3.0 mechanism

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Maximum burning velocity (m/s)	3.15	0.385	Calculated using Cantera and GRI 3.0 mechanism
Heat of vaporization (J/g)*	454.6	510.4	ISO1), Annex A, Table A.3
Lower flammability limit (% vol. fraction)***	4	5.3	ISO1), Annex B, Table B.2
Upper flammability limit (% vol. fraction)***	77	17.0	ISO1), Annex B, Table B.2
Minimum ignition energy (mJ)***	0.017	0.274	ISO1), Annex B, Table B.2
Auto-ignition temperature***	858	810	ISO1), Annex B, Table B.2
Toxicity	Non	Non	Orange book5)
Temperature at critical point (K)	33.19****	190.55	Hydrogen: ISO1), Annex A, Table A.1 Methane: The Japan Society of Mechanical Engineers, Data Book,

Thermophysical
Properties of Fluids (1983)
Pressure at critical point (kPaA) 1315**** 4595
Hydrogen: ISO1), Annex A,
Table A.1
Methane: The Japan
Society of Mechanical
Engineers, Data Book,
Thermophysical
Properties of Fluids (1983)
Remarks: * At their normal boiling points for comparison purpose.
** At normal temperature and pressure.

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Properties of Fluids (1983)
Remarks: * At their normal boiling points for comparison purpose.
** At normal temperature and pressure.
*** Ignition and combustion properties for air mixtures at 25°C
and 101.3 kPaA.
**** Normal Hydrogen.
3.2 Explanation on respective requirements
3.2.1 Ship type (column 'c')
As the hazard associated with hydrogen cargo is flammability but not toxicity, the ship type is considered 2G.
3.2.2 Independent tank type C required (column 'd')
Independent tank type C is allocated only to dangerous goods of class 2.3 whose vapour density is heavier than air. Independent tank type C is considered not to be required for liquefied hydrogen.

3.2.3 Control of vapour space within cargo tank (column 'e')

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3.2.3 Control of vapour space within cargo tank (column 'e')
Special environment controls such as drying and inerting are generally required for liquid chemical products in consideration of the reactivity of cargo vapour and air. As is the case for LNG, it is considered not to be necessary to apply such requirements for liquefied hydrogen.
3.2.4 Vapour detection (column 'f')
Because hydrogen is flammable and non -toxic, it is appropriate to require Flammable (F) as vapour detection for liquefied hydrogen.
3.2.5 Gauging (column 'g')
On the grounds that Closed (C) gauging is required, in principle, for flammable or toxic cargoes,

such as methane, it is considered to be appropriate to require Closed (C) gauging for hydrogen, taking into account that hydrogen has high ignitability and a wide flammable range in air and that closed gauging is effective to prevent leakage of gases into air.

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that closed gauging is effective to prevent leakage of gases into air.

4 SPECIAL REQUIREMENTS AGAINST HAZARDS OF LIQUEFIED HYDROGEN

4.1 Hazards of liquefied hydrogen to be considered

4.1.1 The hazards related to liquefied hydrogen are low ignition energy , a wide range of flammability limits, low visibility of flames in case of fire, high flame velocity which may lead to the detonation with shockwave, low temperature and liquefaction/solidification of inert gas and constituents of air which may result in an oxygen-enriched atmosphere, high permeation, low viscosity, and hydrogen embrittlement including weld metals. Where vacuum insulation is adopted, due consideration should be given to the possibility of untimely deterioration of insulation properties at the envisaged carriage temperatures of liquid hydrogen. The vacuum insulation evaluation should be specified for the normal range or upper limit of cold vacuum

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insulation evaluation should be specified for the normal range or upper limit of cold vacuum pressure (CVP), and loss of vacuum should be defined with respect to this value. Accordingly, effect of vacuum pressure should be taken into account at the time of design and testing of piping with vacuum insulation. Supporting structure and adjacent hull structure should be designed taking into account the cooling owing to loss of vacuum insulation.

4.1.2 Hydrogen is essentially a mixture of ortho - and para-hydrogen, with an equilibrium concentration of 75% ortho-hydrogen and 25% para-hydrogen at ambient temperature. When liquefied at 20K, there is a slow but continuous transformation of ortho -hydrogen to para-hydrogen. The exothermic conversion of the nuclear spin isomers of hydrogen (ortho- to para-hydrogen) may take place and the effect of the conversion may have an impact on the cooling capacity and relief valve capacity of the vessel's equipment.

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the cooling capacity and relief valve capacity of the vessel's equipment.